

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Dowtherm® A**
Cat No. : **H30347**

Unique Formula Identifier (UFI) **H54C-855M-DX0C-U3Q4**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address

begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

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CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Skin Corrosion/Irritation	Category 2 (H315)
Serious Eye Damage/Eye Irritation	Category 2 (H319)
Specific target organ toxicity - (single exposure)	Category 3 (H335)

Environmental hazards

Acute aquatic toxicity	Category 1 (H400)
Chronic aquatic toxicity	Category 1 (H410)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Warning

Hazard Statements

H315 - Causes skin irritation
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H410 - Very toxic to aquatic life with long lasting effects

Precautionary Statements

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P337 + P313 - If eye irritation persists: Get medical advice/attention
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P312 - Call a POISON CENTER or doctor if you feel unwell
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P332 + P313 - If skin irritation occurs: Get medical advice/attention

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Diphenyl oxide	101-84-8	EEC No. 202-981-2	73.5	Eye Irrit. 2 (H319)

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				Aquatic Chronic 2 (H411)
Biphenyl	92-52-4	EEC No. 202-163-5	26.5	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Biphenyl	-	1	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically.
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Carbon dioxide (CO₂). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

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As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 10

Switzerland - Storage of hazardous substances

Storage class - SC 10/12
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority.

- The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is

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administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Diphenyl oxide	TWA: 7 mg/m ³ (8h) TWA: 1 ppm (8h) STEL: 14 mg/m ³ (15min) STEL: 2 ppm (15min)	STEL: 2 ppm 15 min STEL: 14 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 7 mg/m ³ 8 hr	TWA / VME: 1 ppm (8 heures). TWA / VME: 7 mg/m ³ (8 heures). STEL / VLCT: 2 ppm. indicative limit STEL / VLCT: 14 mg/m ³ . indicative limit	TWA: 1 ppm 8 uren TWA: 7 mg/m ³ 8 uren STEL: 2 ppm 15 minuten STEL: 14 mg/m ³ 15 minuten	STEL / VLA-EC: 2 ppm (15 minutos). STEL / VLA-EC: 14.2 mg/m ³ (15 minutos). TWA / VLA-ED: 1 ppm (8 horas) TWA / VLA-ED: 7.1 mg/m ³ (8 horas)
Biphenyl			TWA / VME: 0.2 ppm (8 heures). TWA / VME: 1.5 mg/m ³ (8 heures).	TWA: 0.2 ppm 8 uren TWA: 1.3 mg/m ³ 8 uren	TWA / VLA-ED: 0.2 ppm (8 horas) TWA / VLA-ED: 1.3 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Diphenyl oxide	TWA: 7 mg/m ³ 8 ore. Time Weighted Average TWA: 1 ppm 8 ore. Time Weighted Average	TWA: 1 ppm (8 Stunden). AGW - exposure factor 1 TWA: 7.1 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 1 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time TWA: 7.1 mg/m ³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 1 ppm Höhepunkt: 7.1 mg/m ³	STEL: 2 ppm 15 minutos STEL: 14 mg/m ³ 15 minutos TWA: 1 ppm 8 horas TWA: 7 mg/m ³ 8 horas	STEL: 14 mg/m ³ 15 minuten TWA: 7 mg/m ³ 8 uren	TWA: 1 ppm 8 tunteina TWA: 7 mg/m ³ 8 tunteina STEL: 2 ppm 15 minuutteina STEL: 14 mg/m ³ 15 minuutteina
Biphenyl		Haut	TWA: 0.2 ppm 8 horas		TWA: 0.2 ppm 8 tunteina TWA: 1.3 mg/m ³ 8 tunteina STEL: 0.6 ppm 15 minuutteina STEL: 3.8 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Diphenyl oxide	MAK-KZGW: 2 ppm 15 Minuten MAK-KZGW: 14 mg/m ³ 15 Minuten MAK-TMW: 1 ppm 8 Stunden MAK-TMW: 7 mg/m ³ 8 Stunden	TWA: 1 ppm 8 timer TWA: 7 mg/m ³ 8 timer STEL: 14 mg/m ³ 15 minutter STEL: 2 ppm 15 minutter	STEL: 2 ppm 15 Minuten STEL: 14 mg/m ³ 15 Minuten TWA: 1 ppm 8 Stunden TWA: 7 mg/m ³ 8 Stunden	STEL: 14 mg/m ³ 15 minutach TWA: 7 mg/m ³ 8 godzinach	TWA: 1 ppm 8 timer TWA: 7 mg/m ³ 8 timer STEL: 2 ppm 15 minutter. value from the regulation STEL: 14 mg/m ³ 15 minutter. value from the regulation
Biphenyl	Haut MAK-TMW: 0.2 ppm 8 Stunden MAK-TMW: 1 mg/m ³ 8 Stunden	TWA: 0.2 ppm 8 timer TWA: 1 mg/m ³ 8 timer STEL: 0.4 ppm 15 minutter STEL: 2 mg/m ³ 15 minutter	TWA: 0.2 ppm 8 Stunden TWA: 1.3 mg/m ³ 8 Stunden	STEL: 2 mg/m ³ 15 minutach TWA: 1 mg/m ³ 8 godzinach	TWA: 0.2 ppm 8 timer TWA: 1 mg/m ³ 8 timer STEL: 3 mg/m ³ 15 minutter. value calculated STEL: 0.6 ppm 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Diphenyl oxide	TWA: 7 mg/m ³ TWA: 1 ppm STEL : 14 mg/m ³ STEL : 2 ppm	TWA-GVI: 1 ppm 8 satima. TWA-GVI: 7 mg/m ³ 8 satima. STEL-KGVI: 2 ppm 15 minutama. STEL-KGVI: 14 mg/m ³	TWA: 1 ppm 8 hr. vapour TWA: 7 mg/m ³ 8 hr. vapour STEL: 2 ppm 15 min STEL: 14 mg/m ³ 15 min	STEL: 14 mg/m ³ STEL: 2 ppm TWA: 7 mg/m ³ TWA: 1 ppm	TWA: 5 mg/m ³ 8 hodinách. Ceiling: 10 mg/m ³

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		15 minutama.			
Biphenyl		kože TWA-GVI: 0.2 ppm 8 satima. TWA-GVI: 1.3 mg/m ³ 8 satima.	TWA: 0.2 ppm 8 hr. TWA: 1.5 mg/m ³ 8 hr. STEL: 0.6 ppm 15 min STEL: 4.5 mg/m ³ 15 min		TWA: 1 mg/m ³ 8 hodinách. Potential for cutaneous absorption Ceiling: 3 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Diphenyl oxide	TWA: 1 ppm 8 tundides. TWA: 7 mg/m ³ 8 tundides. STEL: 2 ppm 15 minutites. STEL: 14 mg/m ³ 15 minutites.	TWA: 7 mg/m ³ 8 hr TWA: 1 ppm 8 hr STEL: 14 mg/m ³ 15 min STEL: 200 ppm 15 min	STEL: 2 ppm STEL: 14 mg/m ³ TWA: 1 ppm TWA: 7 mg/m ³	STEL: 14 mg/m ³ 15 percekben. CK TWA: 7 mg/m ³ 8 órában. AK	STEL: 2 ppm STEL: 14 mg/m ³ TWA: 1 ppm 8 klukkustundum. TWA: 7 mg/m ³ 8 klukkustundum.
Biphenyl	TWA: 0.2 ppm 8 tundides. TWA: 1.3 mg/m ³ 8 tundides. STEL: 0.4 ppm 15 minutites. STEL: 2.5 mg/m ³ 15 minutites.		STEL: 0.6 ppm STEL: 4 mg/m ³ TWA: 0.25 ppm TWA: 1.5 mg/m ³	TWA: 1 mg/m ³ 8 órában. AK lehetséges borón keresztül felszívódás	TWA: 0.2 ppm 8 klukkustundum. TWA: 1.3 mg/m ³ 8 klukkustundum. Ceiling: 0.4 ppm Ceiling: 2.6 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Diphenyl oxide	STEL: 14 mg/m ³ STEL: 2 ppm TWA: 7 mg/m ³ TWA: 1 ppm	TWA: 7 mg/m ³ IPRD TWA: 1 ppm IPRD STEL: 14 mg/m ³ STEL: 2 ppm	TWA: 7 mg/m ³ 8 Stunden TWA: 1 ppm 8 Stunden STEL: 14 mg/m ³ 15 Minuten STEL: 2 ppm 15 Minuten	TWA: 1 ppm TWA: 7 mg/m ³ STEL: 2 ppm 15 minuti STEL: 14 mg/m ³ 15 minuti	TWA: 0.7 ppm 8 ore TWA: 5 mg/m ³ 8 ore STEL: 1.4 ppm 15 minute STEL: 10 mg/m ³ 15 minute
Biphenyl	TWA: 10 mg/m ³	TWA: 0.2 ppm IPRD TWA: 1.3 mg/m ³ IPRD STEL: 0.4 ppm STEL: 2.5 mg/m ³			

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Diphenyl oxide	MAC: 5 mg/m ³	Ceiling: 7.1 mg/m ³ TWA: 1 ppm TWA: 7 mg/m ³	TWA: 1 ppm 8 urah TWA: 7 mg/m ³ 8 urah STEL: 14 mg/m ³ 15 minutah STEL: 2 ppm 15 minutah	Binding STEL: 2 ppm 15 minuter Binding STEL: 14 mg/m ³ 15 minuter TLV: 1 ppm 8 timmar. NGV TLV: 7 mg/m ³ 8 timmar. NGV	
Biphenyl		Potential for cutaneous absorption TWA: 0.16 ppm TWA: 1 mg/m ³		Indicative STEL: 0.4 ppm 15 minuter Indicative STEL: 2.5 mg/m ³ 15 minuter TLV: 0.2 ppm 8 timmar. NGV TLV: 1.3 mg/m ³ 8 timmar. NGV	

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

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See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Diphenyl oxide 101-84-8 (73.5)				DNEL = 25mg/kg bw/day
Biphenyl 92-52-4 (26.5)				DNEL = 63mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Diphenyl oxide 101-84-8 (73.5)	DNEL = 14mg/m ³		DNEL = 7mg/m ³	DNEL = 59mg/m ³
Biphenyl 92-52-4 (26.5)				DNEL = 11.17mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Biphenyl 92-52-4 (26.5)	PNEC = 0.017mg/L	PNEC = 2.69mg/kg sediment dw	PNEC = 0.17mg/L	PNEC = 10mg/L	PNEC = 0.528mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Biphenyl 92-52-4 (26.5)	PNEC = 0.0017mg/L	PNEC = 0.269mg/kg sediment dw		PNEC = 16.7mg/kg food	

8.2. Exposure controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	480 minutes	0.3 mm	EN 374	(minimum requirement)

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

In case of insufficient ventilation, wear suitable respiratory equipment

Recommended Filter type: Multi-purpose/ABEK conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure

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limits are exceeded or if irritation or other symptoms are experienced.
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance		
Odor	Unpleasant	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	257 °C / 494.6 °F	
Flammability (liquid)	No data available	
Flammability (solid,gas)	No information available	
Explosion Limits	Lower 0.6% Upper 15.0%	
Flash Point	113 °C / 235.4 °F	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	Not applicable	Solid
Water Solubility	Insoluble in water	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Diphenyl oxide	4.21	
Biphenyl	4.008	
Vapor Pressure	<=1100 hPa @ 50 °C	
Density / Specific Gravity	No data available	
Bulk Density	No data available	
Vapor Density	Not applicable	Solid
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Evaporation Rate Not applicable - Solid

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity None known, based on information available

10.2. Chemical stability Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid Incompatible products. Excess heat.

10.5. Incompatible materials

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Oxidizing agent.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Diphenyl oxide	LD50 = 2450 mg/kg (Rat)	LD50 > 7940 mg/kg (Rabbit)	-
Biphenyl	2140 mg/kg (Rat)	>5010 mg/kg (Rabbit)	-

(b) skin corrosion/irritation;

Category 2

(c) serious eye damage/irritation;

Category 2

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity;

No data available

(f) carcinogenicity;

No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity;

No data available

(h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

Not applicable

Solid

Symptoms / effects, both acute and delayed
No information available.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

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SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Diphenyl oxide	LC50: 4 - 7.9 mg/L, 96h static (Pimephales promelas) LC50: = 4 mg/L, 96h flow-through (Pimephales promelas)	LC50: 0.11 - 1.1 mg/L, 48h (Daphnia magna)	
Biphenyl	LC50: 4.3 - 5.1 mg/L, 96h static (Lepomis macrochirus) LC50: 1.17 - 1.81 mg/L, 96h static (Pimephales promelas) LC50: 1.65 - 2.29 mg/L, 96h flow-through (Pimephales promelas) LC50: 1.4 - 1.6 mg/L, 96h static (Oncorhynchus mykiss)	EC50: 0.63 - 0.85 mg/L, 48h Static (Daphnia magna)	

Component	Microtox	M-Factor
Diphenyl oxide	EC50 = 3.64 mg/L 30 min	
Biphenyl	EC50 = 1.89 mg/L 30 min EC50 = 3.20 mg/L 5 min EC50 = 3.30 mg/L 15 min	1

12.2. Persistence and degradability

Persistence

Insoluble in water.

Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

May have some potential to bioaccumulate

Component	log Pow	Bioconcentration factor (BCF)
Diphenyl oxide	4.21	470 dimensionless
Biphenyl	4.008	1900 dimensionless

12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

Component	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances
Biphenyl	Group III Chemical	

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

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SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

<u>14.1. UN number</u>	UN3082
<u>14.2. UN proper shipping name</u> Technical Shipping Name	Environmentally hazardous substances, liquid, n.o.s. (Biphenyl, Diphenyl ether)
<u>14.3. Transport hazard class(es)</u>	9
<u>14.4. Packing group</u>	III

ADR

<u>14.1. UN number</u>	UN3082
<u>14.2. UN proper shipping name</u> Technical Shipping Name	Environmentally hazardous substances, liquid, n.o.s. (Biphenyl, Diphenyl ether)
<u>14.3. Transport hazard class(es)</u>	9
<u>14.4. Packing group</u>	III

IATA

<u>14.1. UN number</u>	UN3082
<u>14.2. UN proper shipping name</u> Technical Shipping Name	Environmentally hazardous substances, liquid, n.o.s. (Biphenyl, Diphenyl ether)
<u>14.3. Transport hazard class(es)</u>	9
<u>14.4. Packing group</u>	III

<u>14.5. Environmental hazards</u>	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
<u>14.6. Special precautions for user</u>	No special precautions required.
<u>14.7. Maritime transport in bulk according to IMO instruments</u>	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

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International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Diphenyl oxide	101-84-8	202-981-2	-	-	X	X	KE-27676	X	X
Biphenyl	92-52-4	202-163-5	-	-	X	X	KE-02861	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Diphenyl oxide	101-84-8	X	ACTIVE	X	-	X	X	X
Biphenyl	92-52-4	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Diphenyl oxide	101-84-8	-	-	-
Biphenyl	92-52-4	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Diphenyl oxide	101-84-8	Not applicable	Not applicable
Biphenyl	92-52-4	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Diphenyl oxide	WGK2	Class I : 20 mg/m ³ (Massenkonzentration)
Biphenyl	WGK2	Class I : 20 mg/m ³ (Massenkonzentration)

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Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Diphenyl oxide 101-84-8 (73.5)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

H411 - Toxic to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical incident response training.

Prepared By

Revision Date

Revision Summary

Health, Safety and Environmental Department

01-Mar-2024

New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

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COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

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End of Safety Data Sheet